



Phase 1: Strategic Domain Selection

Week 4 Update

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Outline

- Objective & Pipeline
- Recap
- Experimental Results
- Next Steps

Objective

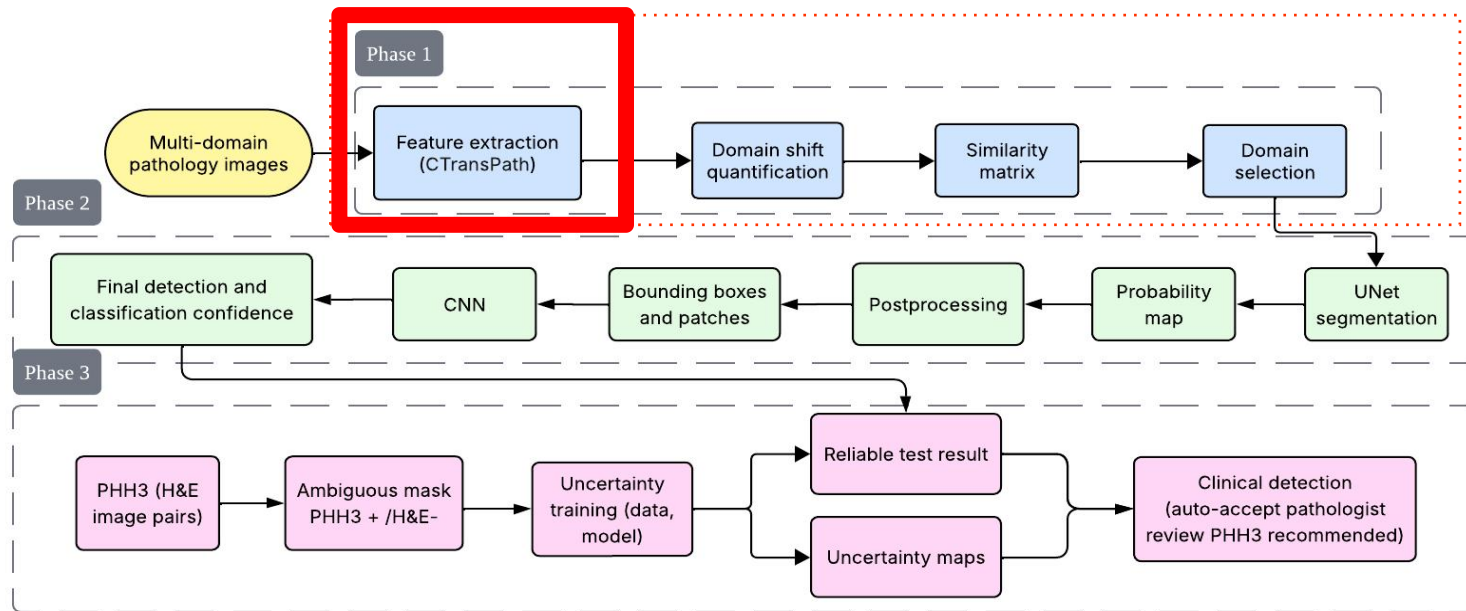
To develop a systematic pipeline to analyze domain relationships, select optimal training domains, and build a robust, ambiguity-aware detection model.

Phase 1: Quantitative Domain Analysis & Strategic Selection

Phase 2: U-Net segmentation & Classification Pipeline

Phase 3: PHH3 Ambiguity Modeling & Uncertainty Quantification

Pipeline



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Recap

Feature Extraction using CTransPath

Patch Extractions of WSIs → CTransPath feature vectors → UMAP visualization

Recap (Next Steps)

Step 1:

- Normalize the slides to flatten distribution
- Redo the CTransPath and UMAP to see if clusters are closer together
 - see if scanner and lab origin clusters reduce in distance
 - better visualize morphological features

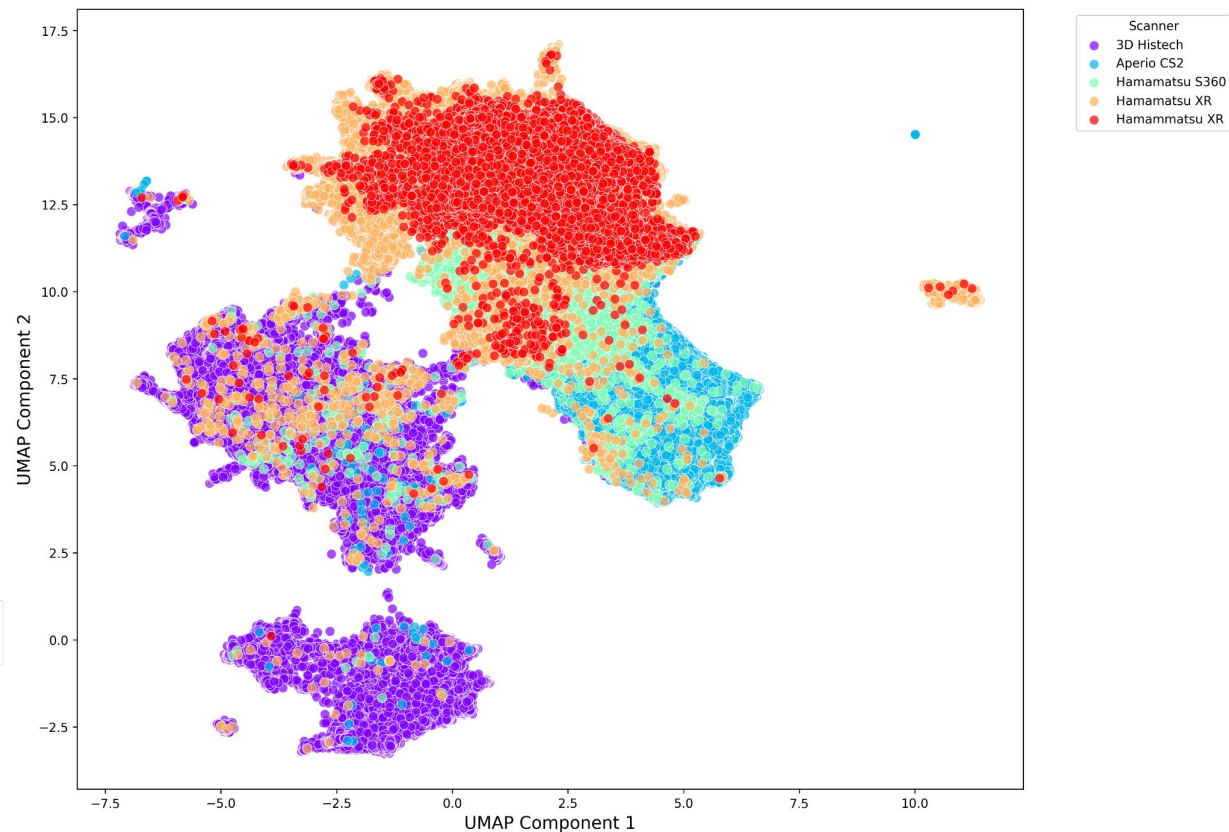
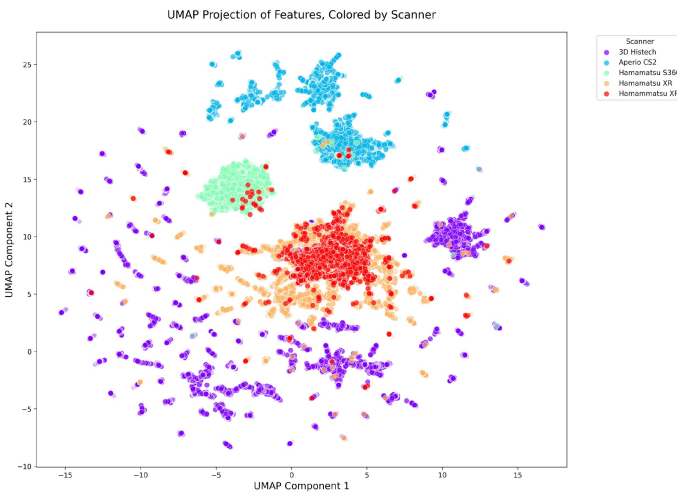
Step 2:

- Feature extraction on unbiased model (EfficientNet)
- Repeat with stain-normalized slides

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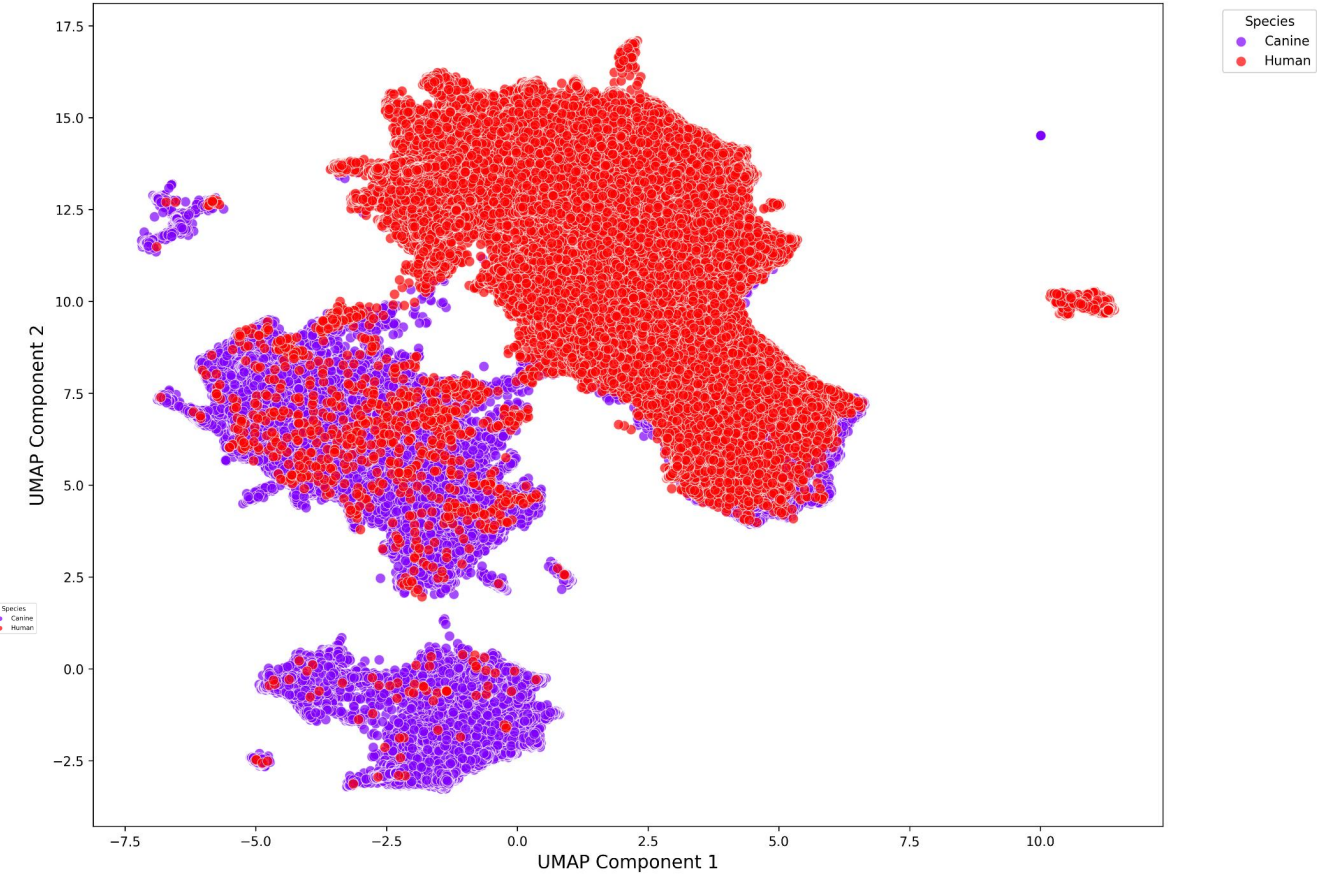
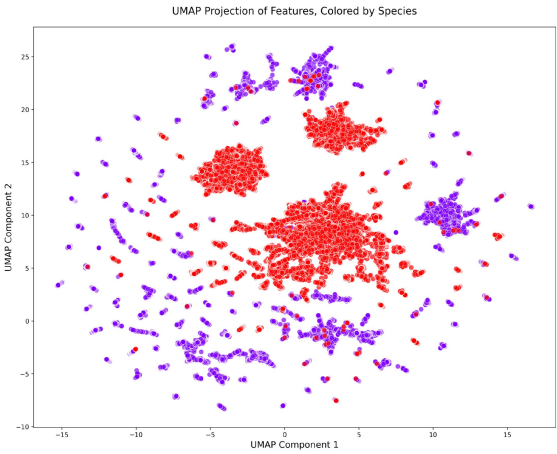
UMAP: Scanner CTransPath vs Efficient Net Bo



UMAP: Species

CTransPath vs Efficient

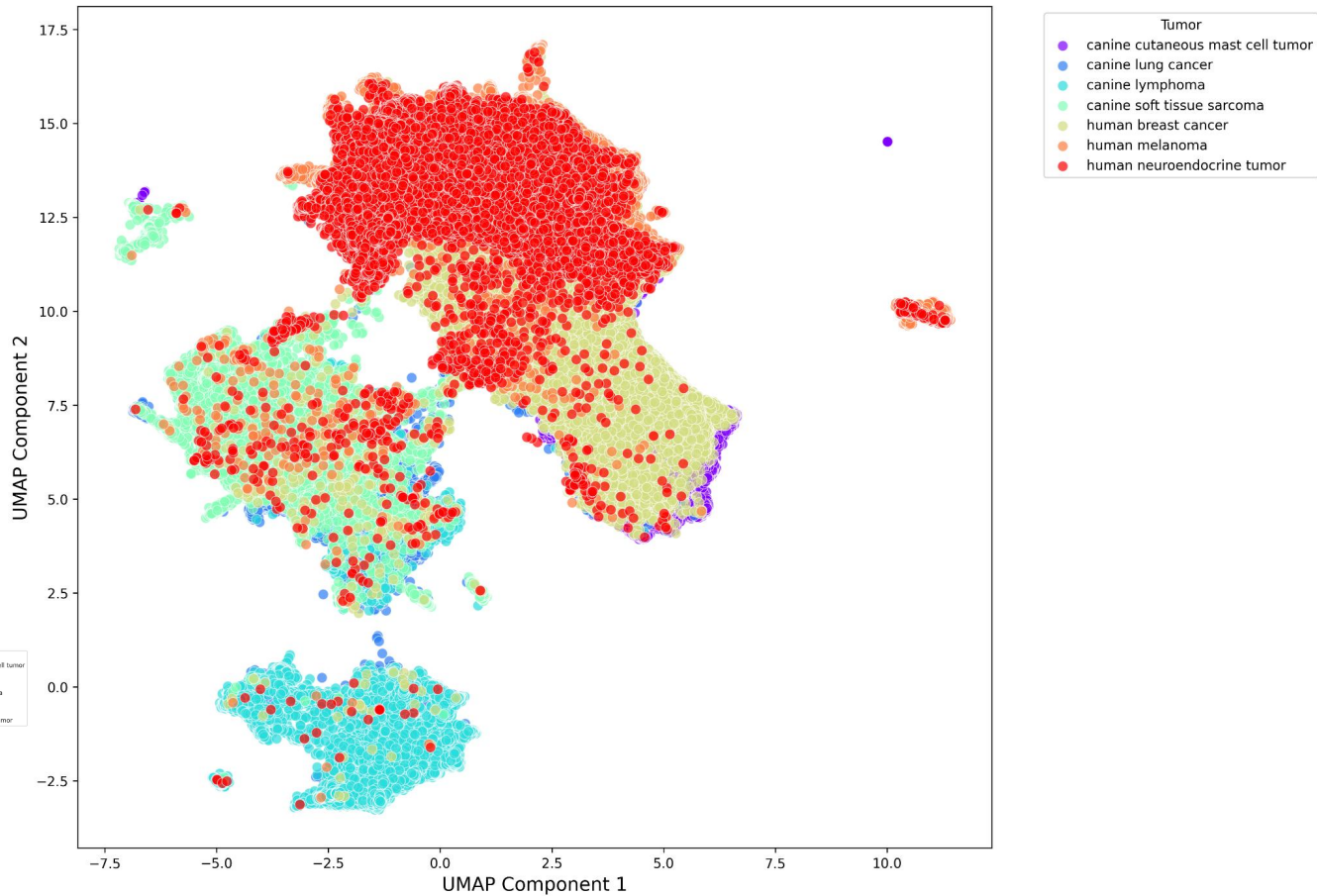
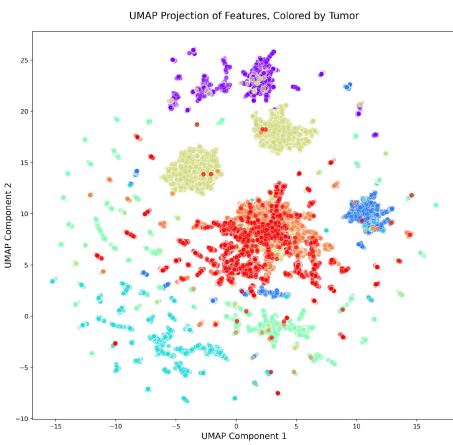
Net Bo



UMAP: Tumor Type

CTransPath vs

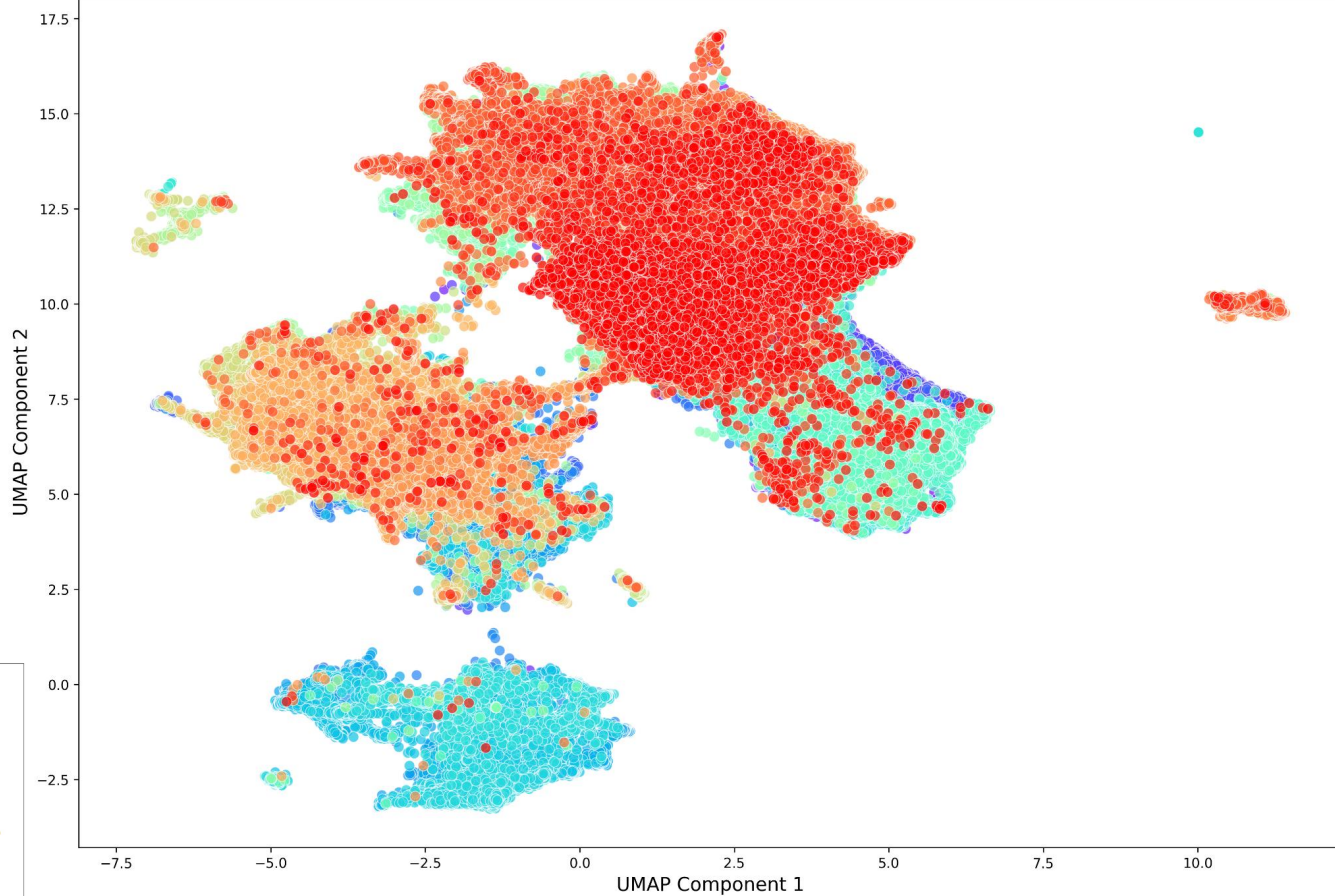
Efficient Net Bo



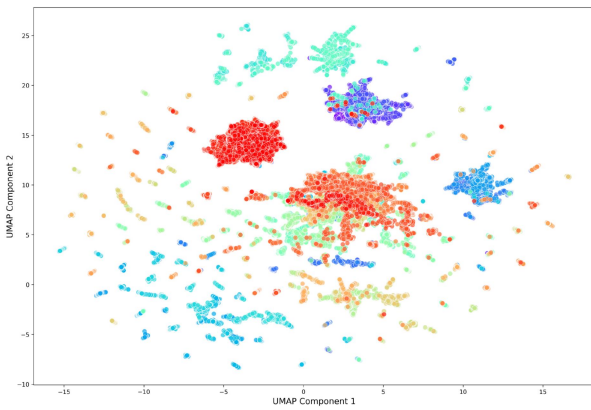
UMAP: WSI

CTransPath vs

Efficient Net Bo



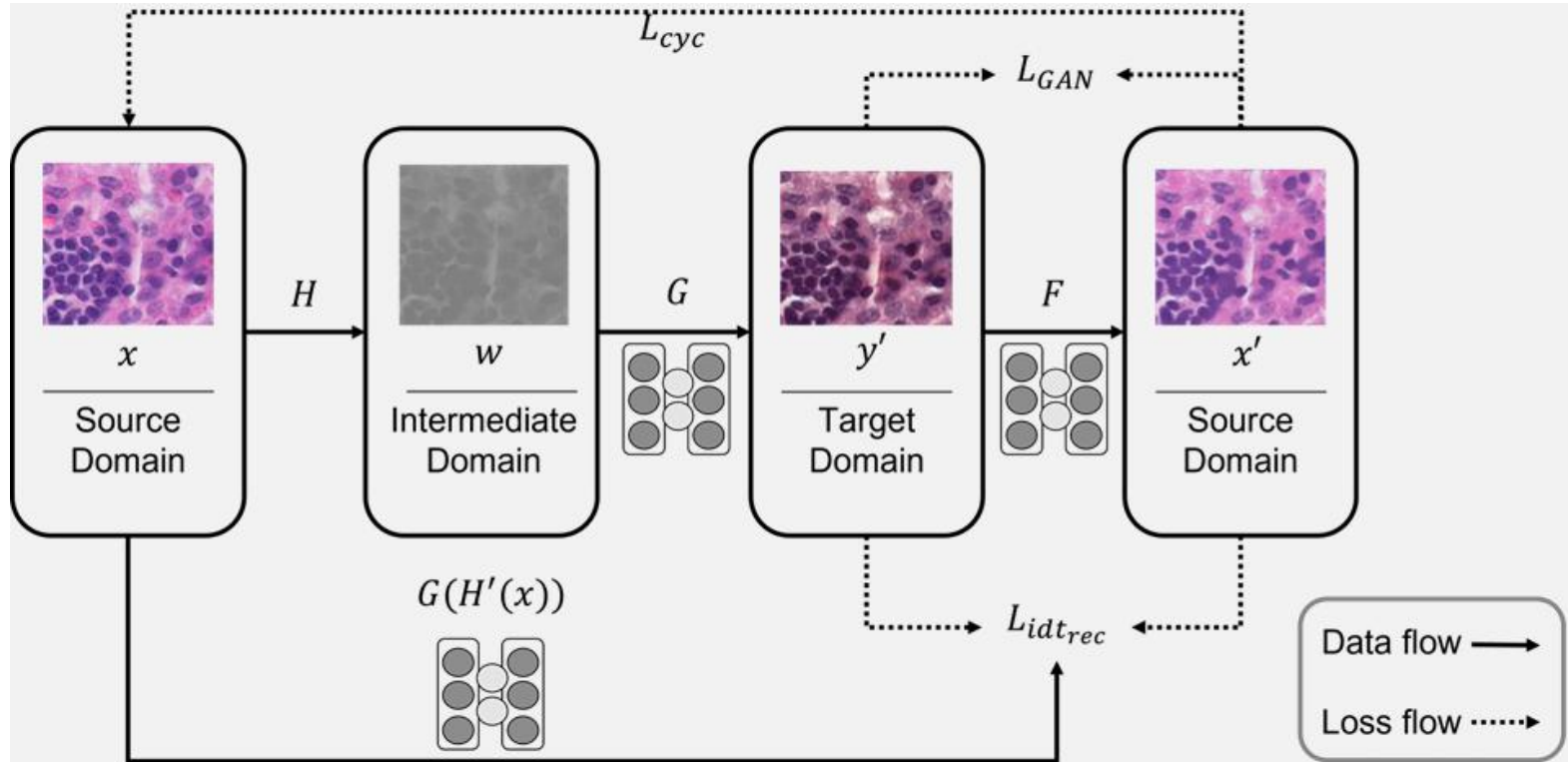
UMAP Projection of Features, Colored by Slide



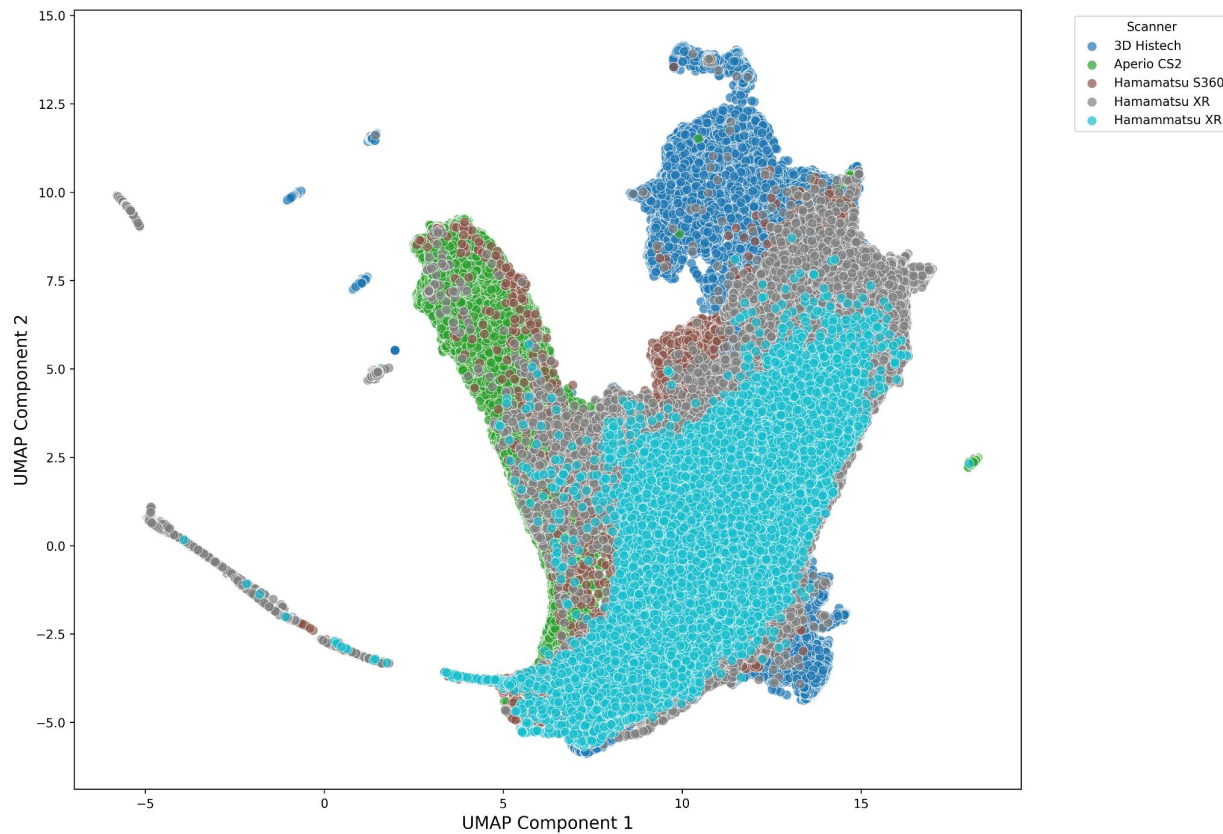
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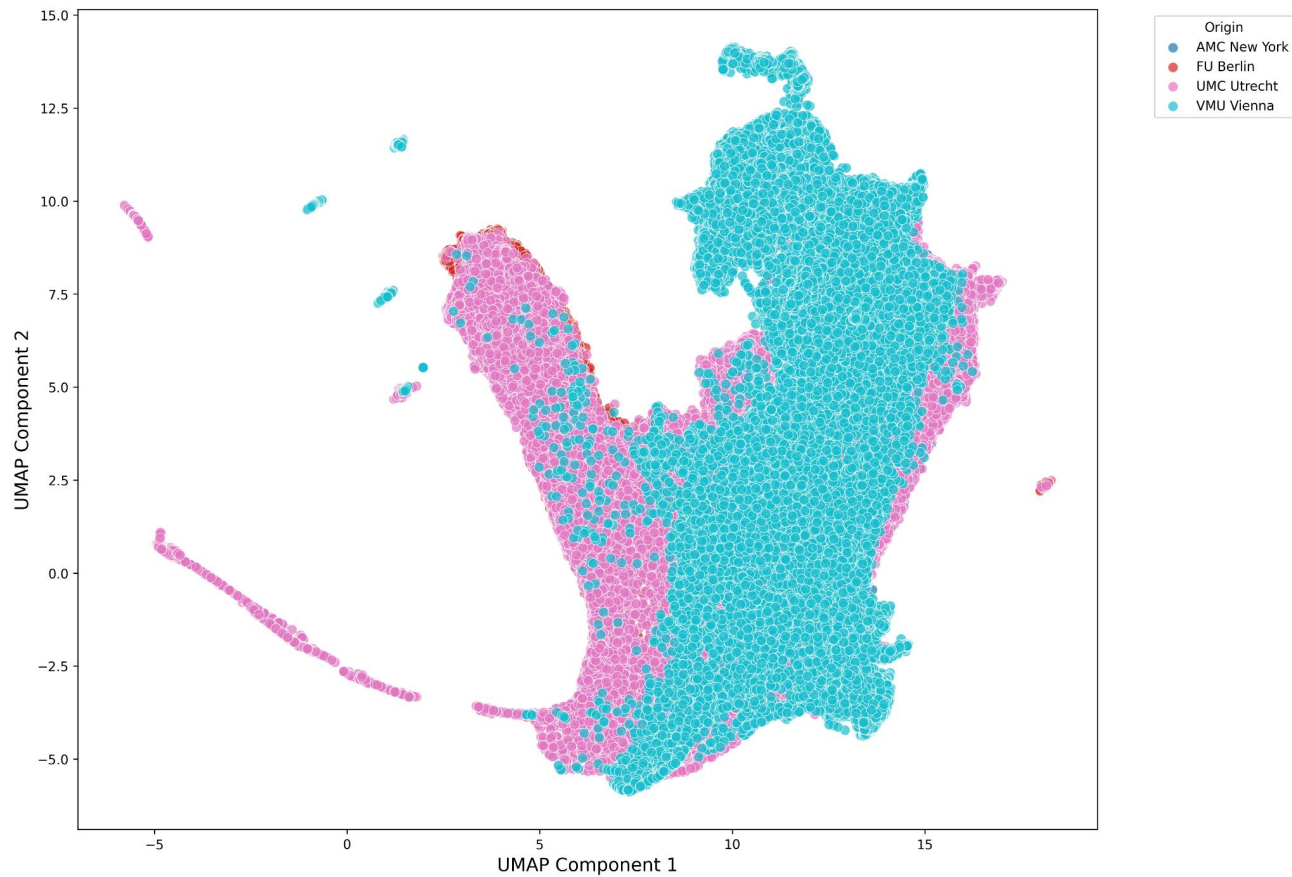
Cycle GAN (Generative Adversarial Network) Stain Normalization



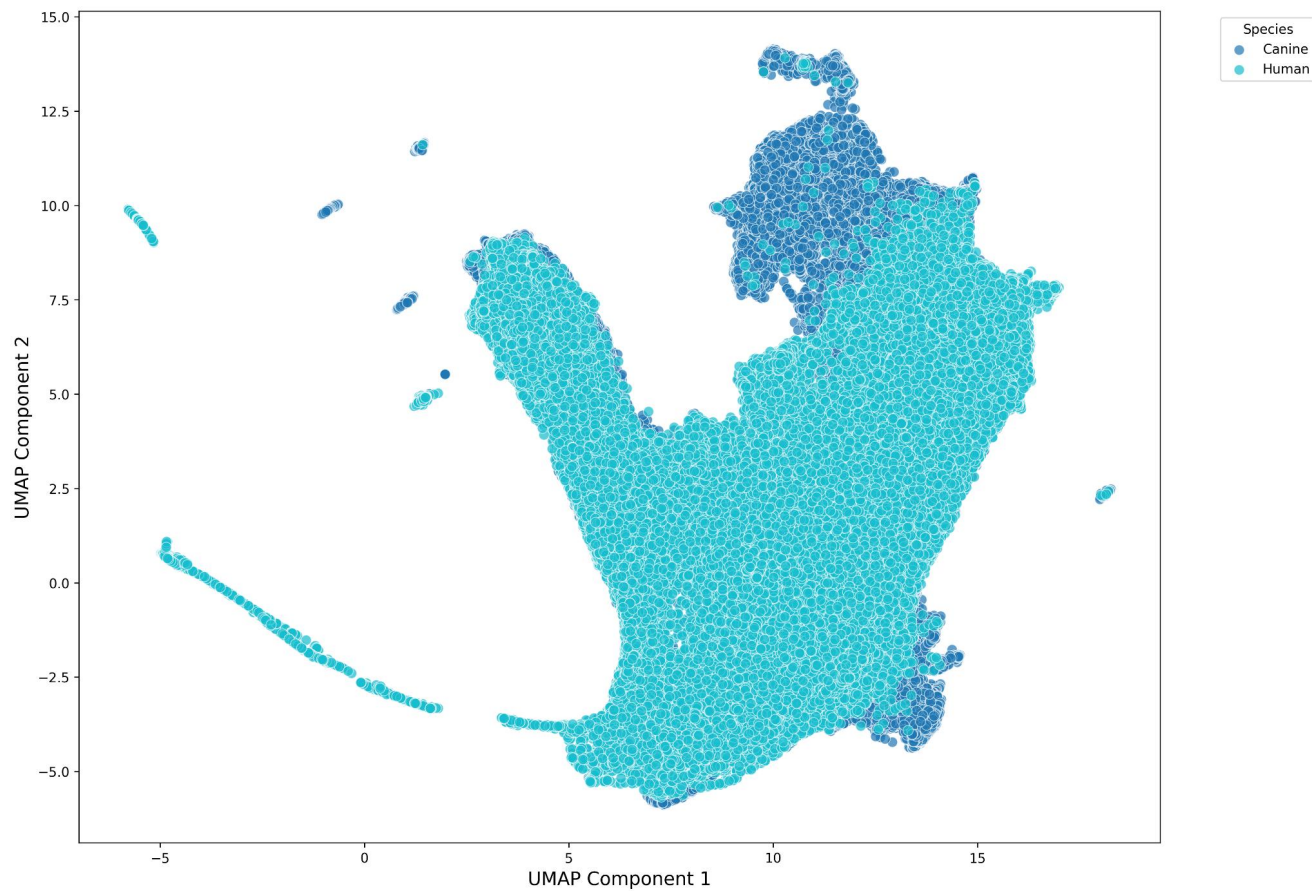
UMAP: Scanner (Cycle GAN + CTransPath)



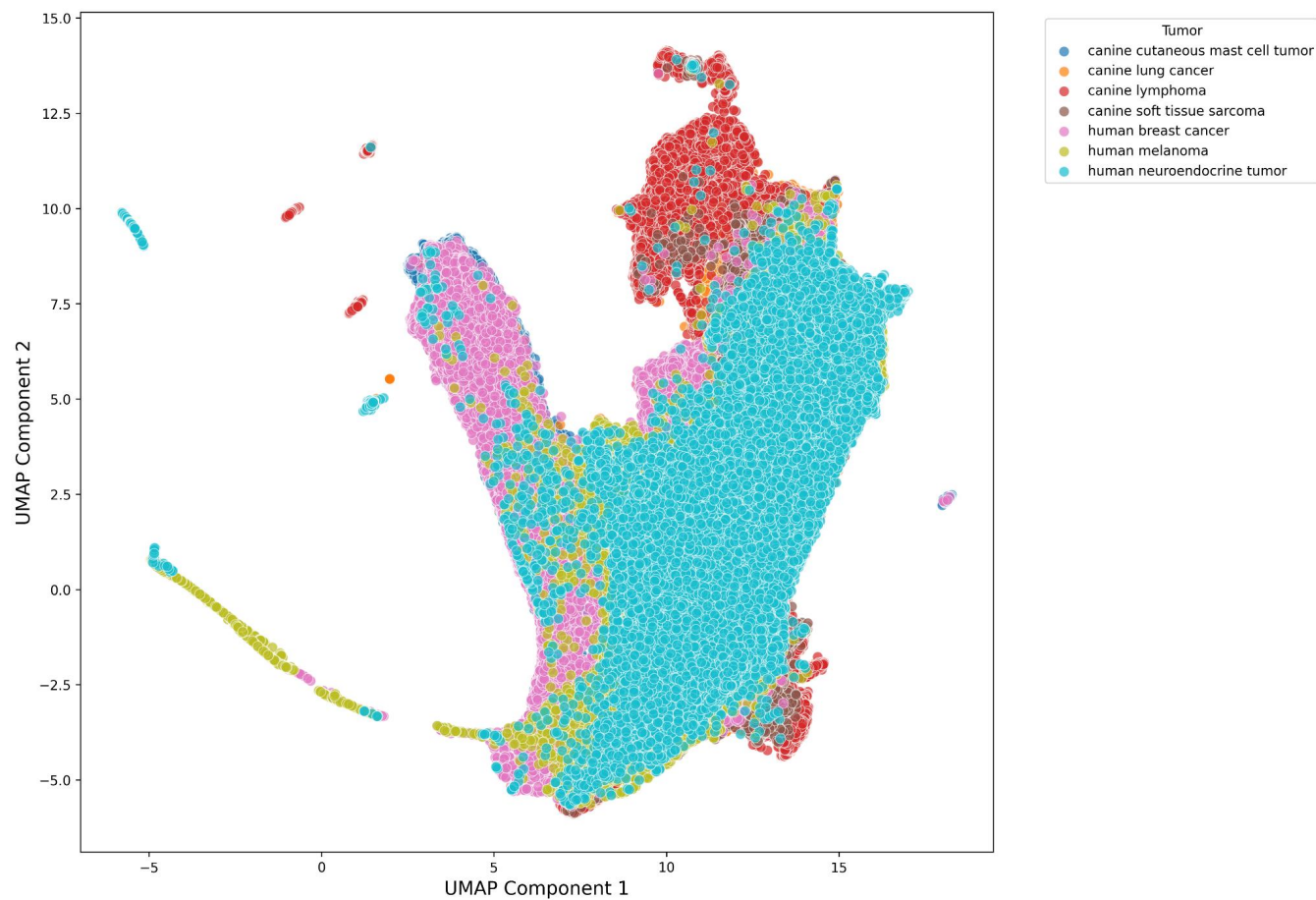
UMAP: Lab Origin (Cycle GAN + CTransPath)



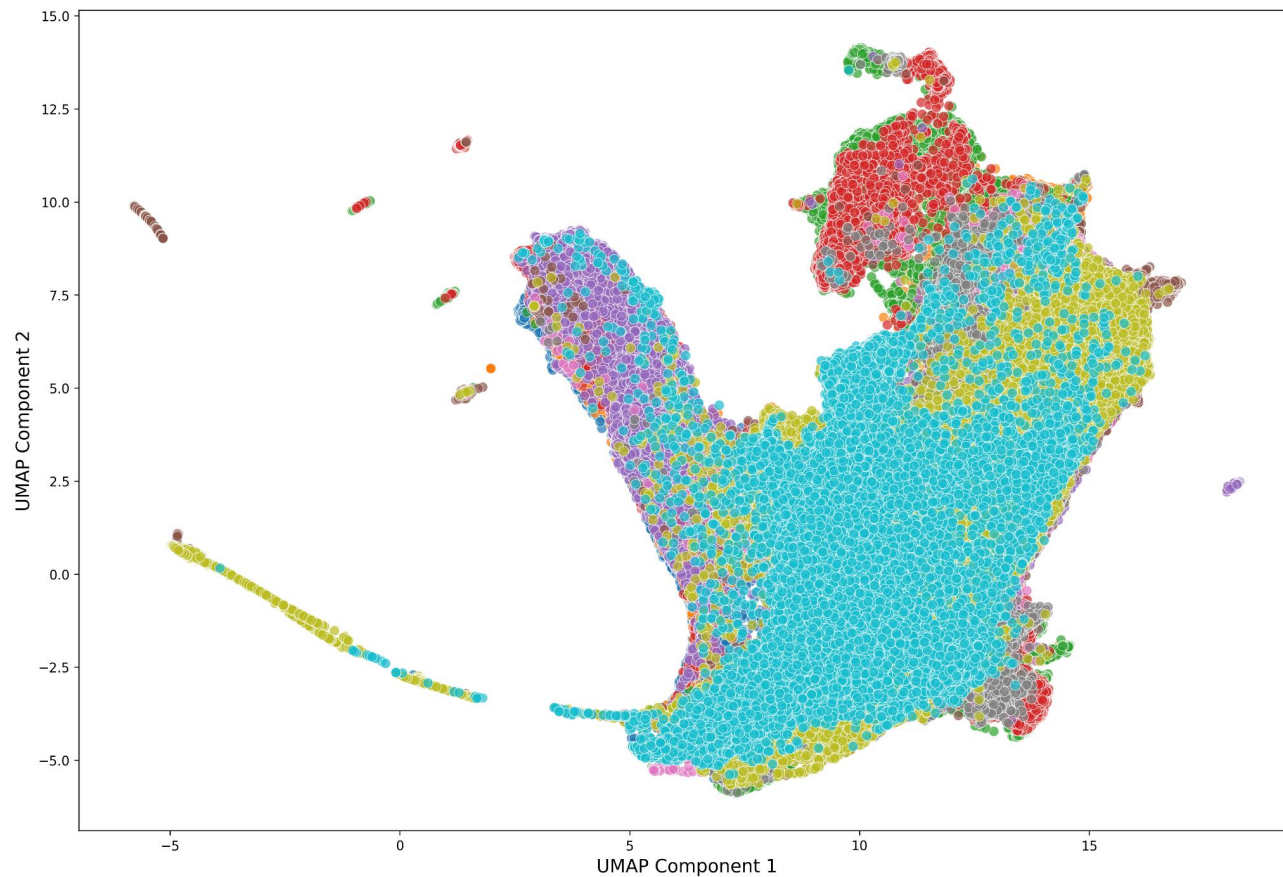
UMAP: Species (Cycle GAN + CTransPath)



UMAP: Tumor Type (Cycle GAN + CTransPath)



UMAP: WSI (Cycle GAN + CTransPath)



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Next Steps (Upcoming 2-4 Weeks)

Step 1:

- Work with other feature extractors to see if we get better results
 - o DINOv3
 - o Mask autoencoder
- Determine if there are natural clusters within the feature space (potential areas to start training data selection)
- Reaching out for PHH3 ground-truth dataset

Step 2:

- Domain shift quantification
 - determine an algorithm for weights on each measurement parameter

Thank you!